DENMARK BUSHFIRE BRIGADES WAERN RADIO USE

# WAERN Radios

Channels, Talk-Around and your position.

RADIO USER GUIDE


1 WAERN Radios
2 Talking on the Radio
3 Common Radio Issues


## Tait vehicle radio

Every vehicle has the yellow head, which runs a VHF high-band and a UHF radio. Where a yellow handset with a display replaces the black one, it adds a second VHF: three radios in all.




Head: VHF + UHF



Handset: 2nd VHF

| | |
|---|---|
| **Change channel** | Key the number on the handset, then **Select**. Or ▲ ▼ to scroll, then **Done**. |
| **Swap VHF / UHF** | F3 BAND. The large channel on the display is the radio you talk on. |
| **Hear both radios** | F1 DUAL MODE once: both channels show, the one you talk on in large type. A second press transmits on both and shows ⇶. Hold F1 for 3 seconds to go back to one radio. |
| **Talk-Around** | F4 TALK AROUND: ⇥‖ shows top left. F4 again, or change channel, to cancel. Only when the IC directs, or if you declare an emergency and get no reply: then ask to be relayed. |
| **Your location** | If asked, describe where you are. Give GPS coordinates only if they're requested: **Menu** › [GPS menu item], ▲ ▼ for latitude/longitude or Zone 50 Easting/Northing. |
| **Scan** | F2 SCAN. Keep it off at incidents unless the IC directs. |

## Vertex VX-829 handheld

Black, VHF high band, yellow aerial.




| | |
|---|---|
| **Key lock** | Long press the orange button on top to lock or unlock the keys. |
| **Zone knob** | 1 DEC/PaW 501–553 · 2 WAERN 101–353 · 3 WAERN simplex 354–369 · 4 WAERN 371–400 · 5 DEC/PaW 601–653. |
| **A Talk-Around** | Only when the IC directs, or if you declare an emergency and get no reply: then ask to be relayed. |
| **B Power** | Normally left as set. |
| **▲ ▼ C Channel** | Side keys, or C then the three-digit channel (zones 2 to 4). Change zone first, or after an off/on it may go to an unintended channel. |
| **D Scan** | Keep it off at incidents unless the IC directs. |
| **Your location** | No GPS. Describe where you are; give coordinates only if requested. |
| **Light on top** | Red: transmitting · green flash: busy · yellow flash: selcall. |

**Use it outdoors**, away from the vehicle and obstacles. Not indoors, and not for air operations or as ground controller. On 173 it may hear the repeater but not reach it.

## Denmark channels

Tait vehicle radio · VX 3 handheld zone · UHF CB appliance crew

| | | | | | |
|---|---|---|---|---|---|
| **173** | Incident command | Tait | **525** | Aerial suppression | Tait |
| **360** | Initial command → Sector 1 | Tait VX 3 | **ESC 220** | Rescue helicopter | Tait UHF |
| **361 · 362** | Sector command | Tait VX 3 | **CB 1–80** | Farm, graders, tankers, and talk within your crew | Tait UHF UHF CB |
| **358 · 359 · 367** | Extra sectors | Tait VX 3 | | | |

Sources: DFES aide-memoires and channel guide, Shire of Denmark preformed plan. Tait keys as reprogrammed.

DENMARK BUSHFIRE BRIGADES WAERN RADIO USE

# Talking on the Radio

Call signs, prowords and the phonetic alphabet.


RADIO USER GUIDE

1 WAERN Radios

2 Talking on the Radio

3 Common Radio Issues


## Making a call

1. **Listen first.** Wait until the channel is clear and you've heard OUT.
2. **Press PTT, wait one second,** then talk.
3. **Call them,** then say THIS IS and your call sign.

**OVER** My turn is finished. A reply is needed.

**OUT** Finished, no reply. Only one says it.

| | |
|---|---|
| YOU | "Denmark Chief, this is Shadforth Light Tanker, over." |
| DENMARK CHIEF | "Shadforth Light Tanker, this is Denmark Chief, go ahead, over." |
| YOU | "Arrived on scene, two crew, starting on the north flank, over." |
| DENMARK CHIEF | "Roger, out." |

**CALL SIGNS** DFES lists when they're needed: your first call, a station joining or leaving, changing channel, closing down, difficult conditions and multi-agency incidents. Otherwise they can be left out, as in the example.

## Moving off the command channel

Keep 173 for command. Crews work on 360, then 361, 362.

1. **"CHANGE TO 360, OVER."** Get ready.
2. **"WILCO, OVER."** Or say if you can't.
3. **"CHANGE NOW, OUT."** Go to 360.
4. **Radio check on 360** straight away.

## Common prowords

| | | | |
|---|---|---|---|
| **WAIT** | Short pause. Others don't transmit. | **WAIT OUT** | Long pause. Others may transmit. |
| **ROGER** | Message understood. Not a "yes". | **WILCO** | Understood, will comply. |
| **SAY AGAIN** | Repeat your last message. | **CORRECTION** | Error. The right version follows. |
| **GO AHEAD** | Send your message. | **SITREP** | What is your situation? |

## Phonetic alphabet

Only in poor conditions, for hard-to-catch words, and for critical details such as Forest Grid references. To spell a word, say it, then I SPELL.

| | | | | | | |
|---|---|---|---|---|---|---|
| **A** Alpha | **B** Bravo | **C** Charlie | **D** Delta | **E** Echo | **F** Foxtrot | **G** Golf |
| **H** Hotel | **I** India | **J** Juliett | **K** Kilo | **L** Lima | **M** Mike | **N** November |
| **O** Oscar | **P** Papa | **Q** Quebec | **R** Romeo | **S** Sierra | **T** Tango | **U** Uniform |
| **V** Victor | **W** Whiskey | **X** X-ray | **Y** Yankee | **Z** Zulu | | |

**FIGURES** **0** zero · **1** wun · **2** too · **3** thuh-ree · **4** fow-wer · **5** fi-yiv · **6** six · **7** se-ven · **8** ate · **9** niner

## Emergency message

Extreme emergency only. Everyone else stops transmitting and listens.

"**EMERGENCY EMERGENCY EMERGENCY**, Denmark Chief, this is Firefighter Smith, Shadforth Light Tanker. Two crew, burnover, sheltering in the appliance at the junction of Turner Rd and Shadforth Rd. Over."

Name and rank, appliance, crew, SITREP, location. On your sector or control channel*.

**AVL:** also hold the Distress Button 4 seconds. ***No reply:** Talk-Around, ask for a relay.


Sources: DFES Voice Procedures aide-memoire, SOP 3.2.4 and SOP 3.5.11; Field Radio Operator manual; ACP 125(G). v1.2 · November 2026

DENMARK BUSHFIRE BRIGADES WAERN RADIO USE

# Common Radio Issues

And how to fix them: ensuring every message is heard.


RADIO USER GUIDE

1 WAERN Radios

2 Talking on the Radio

3 Common Radio Issues

Hear the examples


## 1 Hum on the signal

**Sounds like** a steady low buzz under the voice.

**Common cause:** a handheld transmitting from its charger, indoors.

### IF IT'S YOU

- Lift the handheld out of the charger.
- Step outside, clear of the building.

### IF YOU HEAR IT

> **"Loud, distorted. Hum on your signal, over."**

Say "loud and clear" only when it is.

## 2 Two stations at once

**Sounds like** a warble or squeal. Neither caller is readable.

**Happens** after "any station?" and on a busy fireground.

### IF IT'S YOU

- Listen before you transmit.
- No reply? You weren't heard. Call again.

### IF YOU HEAR IT

> **"Two stations at once. Say again, one at a time, over."**

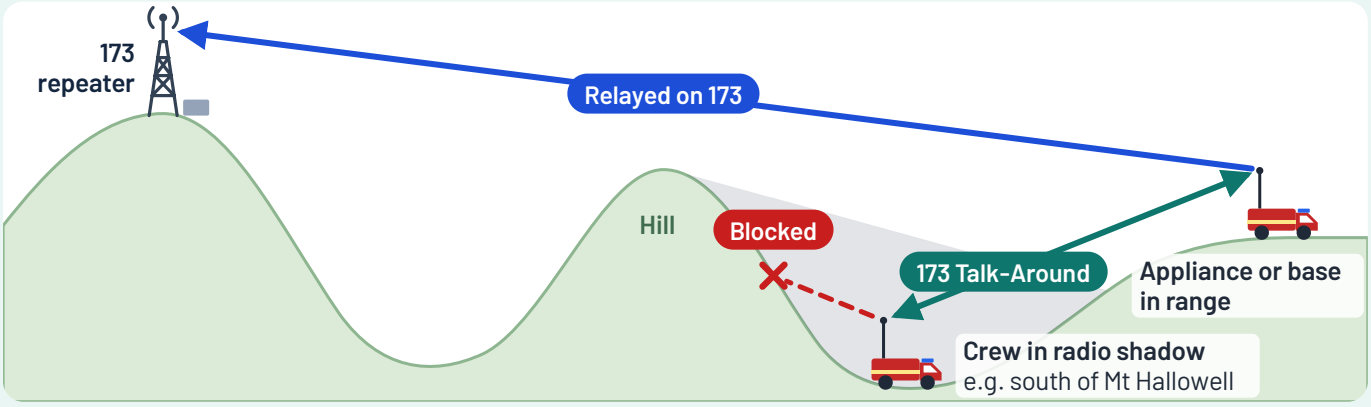
Don't let it pass. Important traffic may be lost.

## Report what you hear

Say "clear" only when it is.

| | | | | | |
|---|---|---|---|---|---|
| STRENGTH | Loud | Good | Weak | Very weak | Fading |
| READABILITY | Clear | Readable | Distorted | Interference | Unreadable |

## 3 Radio shadow? Use Talk-Around




1. **No reply on 173?** You may be in a radio shadow.
2. **Switch to 173 Talk-Around** and call. Say you're on Talk-Around so they reply on it. Radios on 173 in range hear you direct.
3. **Whoever hears you** answers on Talk-Around, then relays on 173.
4. **Switch back to 173** once you're out of the shadow.